There is no 2-$(15, 5, 2)$ design

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Abstract

We prove that no 2-$(15, 5, 2)$ design exists, allowing labelled repeated blocks. We follow Hall and Connor's embedding argument at block size five and prove each result used from Connor. The hypothetical design extends to a symmetric 2-$(22, 7, 2)$ design. Its incidence matrix would have an integer determinant whose square is $49 \cdot 5^{21}$, a contradiction. We give the finite matrix certificates, their checks, and the Lean statements.

1 The statement and conventions

A 2-(v, k, λ) *design* consists of a finite point set P, a finite set $\mathcal{B}$ of block labels, and a subset $B \subseteq P$ for each label $B \in \mathcal{B}$. We use the same letter for a label and its subset. We require $|P| = v$, $|B| = k$ for every block, and exactly λ block labels containing each pair of distinct points. The definition allows different labels to have the same subset; Lemma 2.3 will exclude this for the parameters considered here. We will write $b = |\mathcal{B}|$ and, when it is constant, r for the number of blocks containing a point. A design is *symmetric* when $b = v$.

Theorem 1.1. *There is no* 2-$(15, 5, 2)$ *design. Equivalently, there is no design with* $(v, k, b, r, \lambda) = (15, 5, 21, 7, 2)$.

This is the nonexistence result used in Remark 5.12 and Example 5.13 of Cameron, Goethals, Seidel and Shult [2]. They cite Hall's book; we will use Hall and Connor [4], with all the necessary arguments included here. Their "derived design" is a *residual design* in current terminology: we delete one block and all its points. We will use the current terminology.

We will work over $\mathbb{Q}$ for the matrix arguments, keeping the incidence matrices integral for the final determinant. We will use functional notation for entries: $M(u, w)$ and $\mathbf{x}(u)$. We write I_U for the identity matrix on U, $J_{U,V}$ for the all-ones matrix with row set U and column set V, and $J_U = J_{U,U}$. We omit index sets when their sizes are determined by multiplication. Inner products are written $(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y}$. All ranks and dimensions are over $\mathbb{Q}$.

Our proof follows HC1–HC6 of `cgss/prerequisites.md`. We will first determine block intersections and construct Connor's matrix. Its nonnegative quadratic form and rank bound give the local clique structure. The family of all six-cliques then supplies the seven new points of the embedding. Finally, we will prove the even-order Bruck–Ryser–Chowla obstruction in the single instance needed here.

2 Counting intersections: HC1 and HC2

Lemma 2.1. *Suppose D is a* 2-$(15, 5, 2)$ *design on P with block labels $\mathcal{B}$. Every point belongs to seven blocks, and $|\mathcal{B}| = 21$.*

Proof. Fix $p \in P$, and let r_p be the number of blocks containing p. Count pairs (q, B) with $q \neq p$ and $p, q \in B$. Counting by B gives $4r_p$; counting by q gives $14 \cdot 2$. Thus $r_p = 7$. Counting incident pairs (p, B) now gives $5|\mathcal{B}| = 15 \cdot 7 = 105$, so $|\mathcal{B}| = 21$. These counts concern block labels, so they also apply when blocks repeat. □

The incidence matrix is $N \in \mathrm{Mat}_{P \times \mathcal{B}}(\mathbb{Z})$, with $N(p, B) = 1$ if $p \in B$ and 0 otherwise. Put $G = N^T N$; then $G(B, B') = |B \cap B'|$.

Lemma 2.2. *Suppose D is a 2-$(15, 5, 2)$ design, with incidence matrix N and $G = N^T N$. Then*

$$NN^T = 5I_P + 2J_P, \qquad N\mathbf{j}_{\mathcal{B}} = 7\mathbf{j}_P, \qquad N^T\mathbf{j}_P = 5\mathbf{j}_{\mathcal{B}}, \tag{1}$$

and

$$G^2 = 5G + 50J_{\mathcal{B}}, \qquad GJ_{\mathcal{B}} = J_{\mathcal{B}}G = 35J_{\mathcal{B}}, \qquad J_{\mathcal{B}}^2 = 21J_{\mathcal{B}}. \tag{2}$$

Proof. The diagonal entries of NN^T count the seven blocks on a point; its off-diagonal entries count the two blocks on a pair of points. The other two identities in (1) are the row and column sums of N. Thus

$$G^2 = N^T(NN^T)N = 5N^TN + 2N^TJ_PN = 5G + 50J_{\mathcal{B}},$$

since every entry of N^TJ_PN is $5 \cdot 5 = 25$. Also $G\mathbf{j}_{\mathcal{B}} = N^T(7\mathbf{j}_P) = 35\mathbf{j}_{\mathcal{B}}$. Symmetry gives the other product with $J_{\mathcal{B}}$. Finally, each entry of $J_{\mathcal{B}}^2$ is $|\mathcal{B}| = 21$. □

Lemma 2.3. [4, Lemma 4.1, at $k = 5$] *Suppose D is a 2-$(15, 5, 2)$ design. Distinct block labels have intersection size one or two. Every block has ten other blocks meeting it in one point, and ten meeting it in two points.*

Proof. Fix B, and for $0 \leq i \leq 5$ let n_i count the other labels B' with $|B \cap B'| = i$. Counting other labels, incident points of B, and ordered pairs of distinct points of B, respectively, gives

$$\sum_{i=0}^{5} n_i = 20, \qquad \sum_{i=0}^{5} in_i = 5(7-1) = 30, \qquad \sum_{i=0}^{5} i(i-1)n_i = 5 \cdot 4(2-1) = 20.$$

For the last equality, every ordered pair of distinct points of B lies in exactly one further block label. Taking this linear combination of the three counts, in $\mathbb{Z}$, gives

$$\sum_{i=0}^{5} (i-1)(i-2)n_i = 20 - 2 \cdot 30 + 2 \cdot 20 = 0.$$

The coefficients at $i = 0, 1, 2, 3, 4, 5$ are $2, 0, 0, 2, 6, 12$. All summands are nonnegative, so $n_i = 0$ unless $i = 1$ or $i = 2$. The first two counts give $n_1 + n_2 = 20$ and $n_1 + 2n_2 = 30$, whence $n_1 = n_2 = 10$. In particular, distinct block labels cannot represent the same subset, since that would give intersection size five. □

We will denote by Z the graph on $\mathcal{B}$ in which distinct blocks are adjacent when their intersection has size one. We write $B \sim B'$ for adjacency and $\Gamma(B)$ for the neighbourhood of B. A *clique* is a set of vertices any two distinct members of which are adjacent.

Lemma 2.4. [4, Lemma 4.2, at $k = 5$] *Suppose D is a 2-$(15, 5, 2)$ design and $B_1 \neq B_2$ are blocks. For $i, j \in \{1, 2\}$ let x_{ij} count the blocks $B \notin \{B_1, B_2\}$ with $|B_1 \cap B| = i$ and $|B_2 \cap B| = j$. The counts are*

$\lvert B_1 \cap B_2 \rvert$	x_{11}	x_{12}	x_{21}	x_{22}
1	5	4	4	6
2	4	6	6	3

In particular, Z has degree ten, and adjacent vertices have five common neighbours.

Proof. Put $s = |B_1 \cap B_2|$. There are 19 remaining block labels. The sum of their intersections with either initial block is $30 - s$, by the preceding proof. By (2), the sum of the products of these two intersections is

$$(G^2)(B_1, B_2) - 5s - 5s = (5s + 50) - 10s = 50 - 5s.$$

Partitioning the remaining labels by their two intersection sizes gives

$$\begin{aligned} x_{11} + x_{12} + x_{21} + x_{22} &= 19, \\ x_{11} + x_{12} + 2x_{21} + 2x_{22} &= 30 - s, \\ x_{11} + 2x_{12} + x_{21} + 2x_{22} &= 30 - s, \\ x_{11} + 2x_{12} + 2x_{21} + 4x_{22} &= 50 - 5s. \end{aligned}$$

Thus

$$\begin{aligned} x_{22} &= (50 - 5s) - 2(30 - s) + 19 = 9 - 3s, \\ x_{12} = x_{21} &= (30 - s) - 19 - x_{22} = 2 + 2s, \\ x_{11} &= 19 - x_{12} - x_{21} - x_{22} = 6 - s. \end{aligned}$$

For example, the first identity sums $(i - 1)(j - 1)$ over the remaining blocks, and the second sums $j - 1$ or $i - 1$. Substituting $s = 1$ and $s = 2$ proves the table. The degree assertion is Lemma 2.3. □

Thus Z is strongly regular: it has 21 vertices, valency ten, five common neighbours for each adjacent pair and four for each nonadjacent pair. We will reserve k for the design's block size. We will not use a classification of graphs with these parameters.

3 Connor's matrix: HC3

Lemma 3.1. *Suppose D is a 2-$(15, 5, 2)$ design with incidence matrix N and $G = N^T N$. Define*

$$C = 35I_{\mathcal{B}} - 7G + 10J_{\mathcal{B}}. \tag{3}$$

Then C is symmetric,

$$C^2 = 35C, \qquad CN^T = 0, \tag{4}$$

and its entries are

$$C(B, B') = \begin{cases} 10, & B = B'; \\ 3, & B \sim B'; \\ -4, & B \neq B' \text{ and } B \not\sim B'. \end{cases} \tag{5}$$

For every $\mathbf{x} \in \mathbb{Q}^{\mathcal{B}}$,

$$35\mathbf{x}^T C\mathbf{x} = (C\mathbf{x})^T (C\mathbf{x}) \geq 0. \tag{6}$$

Moreover, $\operatorname{rk} C \leq 6$.

Proof. Symmetry and the entries follow from the definition and Lemma 2.3. Using (2), we obtain

$$\begin{aligned} C^2 &= 1225I + 49G^2 + 100J^2 - 490G + 700J - 140GJ \\ &= 1225I + 49(5G + 50J) + 2100J - 490G + 700J - 4900J \\ &= 1225I - 245G + 350J = 35C. \end{aligned}$$

Here both G and J are indexed by $\mathcal{B}$. Also $N^T J_P = 5J_{\mathcal{B},P}$ and $J_{\mathcal{B}} N^T = 7J_{\mathcal{B},P}$, so

$$\begin{aligned} CN^T &= 35N^T - 7N^T(NN^T) + 10J_{\mathcal{B}}N^T \\ &= 35N^T - 7N^T(5I_P + 2J_P) + 70J_{\mathcal{B},P} = 0. \end{aligned}$$

The first identity of (4) and symmetry give (6); its right side is a sum of rational squares.

The determinant formula of Lemma 7.1, whose proof is independent of the design argument, gives

$$\det(NN^T) = \det(5I_{15} + 2J_{15}) = 5^{14} \cdot 35 \neq 0.$$

Thus NN^T is invertible and has rank 15. Over $\mathbb{Q}$, the matrix rank identities give $\operatorname{rk} N = \operatorname{rk}(NN^T) = 15$ and $\operatorname{rk} N^T = \operatorname{rk} N$. Since $CN^T = 0$, the rank inequality for a zero product gives

$$\operatorname{rk} C + \operatorname{rk} N^T \leq |\mathcal{B}| = 21.$$

Thus $\operatorname{rk} C \leq 21 - 15 = 6$. The same determinant formula will serve again at $n = 22$ in the final obstruction. □

We will say that a symmetric rational matrix Q is *positive semidefinite* when $\mathbf{x}^T Q \mathbf{x} \geq 0$ for every rational vector $\mathbf{x}$. The following argument gives the quadratic inequality and the principal-submatrix rank bound needed in the core proof.

Lemma 3.2. *Suppose D is a 2-$(15, 5, 2)$ design, and let C be the matrix in* (3)*. For a subset $F \subseteq \mathcal{B}$, let C_F be the principal submatrix indexed by F. Then C_F is positive semidefinite and $\operatorname{rk} C_F \leq 6$. In particular, if $|F| \geq 7$, then C_F is singular.*

Proof. Extend a vector on F by zero outside F and apply (6). This proves positive semidefiniteness. The rank of a submatrix is at most that of the original matrix, so $\operatorname{rk} C_F \leq \operatorname{rk} C \leq 6$. A square matrix of size $|F| \geq 7$ and rank at most six is singular. □

We have proved the part of Connor's Theorem 3.1 that [4] uses for singularity, with the stronger rank bound. The determinant sign assertion is recorded separately below. Connor's paper has not been consulted; these are reconstructions of the statements reported in Hall and Connor.

Remark 3.3. For comparison with [4]; not used below. Every symmetric positive semidefinite rational matrix Q has nonnegative determinant. We prove this by induction on its size; the determinant of the empty matrix is 1. The diagonal entries are nonnegative by testing coordinate vectors. If a diagonal entry, say $Q(1,1)$, is zero, suppose that $Q(1,i) \neq 0$ for some i. For $\tau = -(Q(i,i) + 1)/(2Q(1,i))$, the vector $\tau \mathbf{e}_1 + \mathbf{e}_i$ has quadratic value -1, contradicting positive semidefiniteness. Thus $Q(1,i) = 0$ for every i, the first row is zero and $\det Q = 0$. In the remaining case $d = Q(1,1) > 0$, write

$$Q = \begin{pmatrix} d & \mathbf{u}^T \\ \mathbf{u} & V \end{pmatrix}, \qquad Q' = V - d^{-1}\mathbf{u}\mathbf{u}^T.$$

Testing Q on $(-d^{-1}\mathbf{u}^T\mathbf{y}, \mathbf{y})$ gives $\mathbf{y}^T Q' \mathbf{y} \geq 0$. Subtracting $\mathbf{u}(i)/d$ times the first row from each later row gives $\det Q = d \det Q'$. The induction hypothesis gives $\det Q' \geq 0$, and the result follows. In particular, $\det C_F \geq 0$ for the submatrices in Lemma 3.2.

4 The local restrictions: HC4

Lemma 4.1. *Suppose D is a 2-$(15,5,2)$ design. Its graph Z has no induced claw: no block is adjacent to three distinct, pairwise nonadjacent blocks.*

Proof. On a putative centre followed by its three leaves, the principal submatrix of C would be

$$Q_{\text{claw}} = \begin{pmatrix} 10 & 3 & 3 & 3 \\ 3 & 10 & -4 & -4 \\ 3 & -4 & 10 & -4 \\ 3 & -4 & -4 & 10 \end{pmatrix}.$$

For $\mathbf{x} = (-1,1,1,1)^T$ its quadratic value is $40 + 2(-9-12) = -2$, contradicting Lemma 3.2. □

Remark 4.2. [4, Lemmas 4.3 and 4.4, at $k = 5$] For comparison with Hall–Connor; not used below. Suppose D is a 2-$(15,5,2)$ design. If distinct blocks have either of the following intersection matrices, then $t = 1$:

$$\begin{pmatrix} 5 & 1 & 1 & 1 & 1 & 2 \\ & 5 & 1 & 2 & 2 & 1 \\ & & 5 & 2 & 2 & 1 \\ & & & 5 & t & 2 \\ & & & & 5 & 2 \\ & & & & & 5 \end{pmatrix}, \qquad \begin{pmatrix} 5 & 1 & 1 & 2 & 2 \\ & 5 & 1 & 1 & 1 \\ & & 5 & 1 & 1 \\ & & & 5 & t \\ & & & & 5 \end{pmatrix}.$$

The omitted entries are determined by symmetry, and t is the intersection size of the fourth and fifth blocks.

By Lemma 2.3, $t \in \{1,2\}$. If $t = 2$ in the first matrix, the first block is the centre of a claw with leaves the second, fourth and fifth blocks. If $t = 2$ in the second matrix, the second block is the centre of a claw with leaves the first, fourth and fifth blocks. Both contradict Lemma 4.1.

Lemma 4.3. [4, Lemma 4.5, at $k = 5$] *Suppose D is a 2-$(15,5,2)$ design, and five distinct blocks B_1, B_2, B_3, Y, W have intersection matrix*

$$\begin{pmatrix} 5 & 1 & 1 & 1 & 2 \\ & 5 & 1 & 2 & 1 \\ & & 5 & 2 & 1 \\ & & & 5 & t \\ & & & & 5 \end{pmatrix},$$

with the lower triangle supplied by symmetry. Then $t = 2$.

Proof. If $t = 1$, the principal submatrix of C is

$$Q_{\text{five}} = \begin{pmatrix} 10 & 3 & 3 & 3 & -4 \\ 3 & 10 & 3 & -4 & 3 \\ 3 & 3 & 10 & -4 & 3 \\ 3 & -4 & -4 & 10 & 3 \\ -4 & 3 & 3 & 3 & 10 \end{pmatrix}.$$

For $\mathbf{x} = (-1,1,1,1,-1)^T$, the diagonal contribution is 50 and the sum over the upper off-diagonal triangle is -27. Thus $\mathbf{x}^T Q_{\text{five}} \mathbf{x} = 50 - 54 = -4$, contradicting Lemma 3.2. Lemma 2.3 leaves only $t = 2$. □

We will also need two seven-by-seven matrices. We give integer certificates for their inverses so that the argument does not depend on a determinant expansion or on Connor's computational Lemma 3.1.

Lemma 4.4. *The following rational matrices Q_1, Q_2 are nonsingular:*

$$Q_1 = \begin{pmatrix} 10 & 3 & 3 & 3 & -4 & -4 & -4 \\ 3 & 10 & 3 & -4 & 3 & 3 & 3 \\ 3 & 3 & 10 & -4 & 3 & 3 & 3 \\ 3 & -4 & -4 & 10 & -4 & -4 & -4 \\ -4 & 3 & 3 & -4 & 10 & 3 & 3 \\ -4 & 3 & 3 & -4 & 3 & 10 & 3 \\ -4 & 3 & 3 & -4 & 3 & 3 & 10 \end{pmatrix}, \qquad Q_2 = \begin{pmatrix} 10 & 3 & 3 & 3 & 3 & -4 & -4 \\ 3 & 10 & 3 & -4 & -4 & 3 & 3 \\ 3 & 3 & 10 & -4 & -4 & 3 & 3 \\ 3 & -4 & -4 & 10 & 3 & -4 & -4 \\ 3 & -4 & -4 & 3 & 10 & -4 & -4 \\ -4 & 3 & 3 & -4 & -4 & 10 & 3 \\ -4 & 3 & 3 & -4 & -4 & 3 & 10 \end{pmatrix}.$$

Proof. Put

$$K_1 = \begin{pmatrix} 20 & -12 & -12 & -6 & 8 & 8 & 8 \\ -12 & 9 & 7 & 4 & -5 & -5 & -5 \\ -12 & 7 & 9 & 4 & -5 & -5 & -5 \\ -6 & 4 & 4 & 4 & -2 & -2 & -2 \\ 8 & -5 & -5 & -2 & 5 & 3 & 3 \\ 8 & -5 & -5 & -2 & 3 & 5 & 3 \\ 8 & -5 & -5 & -2 & 3 & 3 & 5 \end{pmatrix}, \qquad K_2 = \begin{pmatrix} 17 & -11 & -11 & -7 & -7 & 6 & 6 \\ -11 & 9 & 7 & 5 & 5 & -4 & -4 \\ -11 & 7 & 9 & 5 & 5 & -4 & -4 \\ -7 & 5 & 5 & 5 & 3 & -2 & -2 \\ -7 & 5 & 5 & 3 & 5 & -2 & -2 \\ 6 & -4 & -4 & -2 & -2 & 4 & 2 \\ 6 & -4 & -4 & -2 & -2 & 2 & 4 \end{pmatrix}.$$

Integer multiplication gives

$$K_1 Q_1 = 14 I_7, \qquad K_2 Q_2 = 14 I_7. \tag{7}$$

These are explicit certificates: for each $u, w \in \{1, \dots, 7\}$, the sum $\sum_{i=1}^{7} K_j(u,i) Q_j(i,w)$ is 14 when $u = w$ and 0 otherwise. For example, the first two entries of $K_1 Q_1$ are

$$200 - 36 - 36 - 18 - 32 - 32 - 32 = 14, \qquad 60 - 120 - 36 + 24 + 24 + 24 + 24 = 0.$$

All entries use this same displayed multiplication; the check in Section 8 reads the matrices directly from this source and checks each entry with integer arithmetic. If $Q_j \mathbf{x} = 0$, then $14\mathbf{x} = K_j Q_j \mathbf{x} = 0$. Since $14 \neq 0$ in $\mathbb{Q}$, $\mathbf{x} = 0$. Thus Q_j is nonsingular. □

Both determinants are $235298 = 2 \cdot 7^6 > 0$. We will use their nonsingularity against the rank bound, not the sign of these determinants.

5 The six-cliques: the local part of HC5

Lemma 5.1. *Suppose D is a 2-$(15, 5, 2)$ design and $B_1 \sim B_2$ in its block graph Z. There are six-cliques S_1, S_2 such that*

$$B_1, B_2 \in S_1, \qquad S_1 \cap S_2 = \{B_1\}, \qquad S_1 \cup S_2 = \{B_1\} \cup \Gamma(B_1).$$

Moreover, S_1 is the unique six-clique containing B_1 and B_2.

Proof. We will partition the blocks other than B_1, B_2 into

$$\begin{aligned}\mathcal{M} &= \Gamma(B_1) \cap \Gamma(B_2),\\ \mathcal{U} &= \Gamma(B_1) \setminus (\Gamma(B_2) \cup \{B_2\}),\\ \mathcal{V} &= \Gamma(B_2) \setminus (\Gamma(B_1) \cup \{B_1\}),\\ \mathcal{R} &= \mathcal{B} \setminus (\{B_1, B_2\} \cup \mathcal{M} \cup \mathcal{U} \cup \mathcal{V}).\end{aligned}$$

By Lemma 2.4, these disjoint sets have sizes $5, 4, 4, 6$, respectively. In particular, $\Gamma(B_1) = \{B_2\} \cup \mathcal{M} \cup \mathcal{U}$ and $\Gamma(B_2) = \{B_1\} \cup \mathcal{M} \cup \mathcal{V}$, as disjoint unions.

Both $\mathcal{U}$ and $\mathcal{V}$ are cliques. For distinct nonadjacent $Y, Y' \in \mathcal{U}$, the block B_1 would be the centre of a claw with leaves B_2, Y, Y'. For distinct nonadjacent $W, W' \in \mathcal{V}$, the centre B_2 and leaves B_1, W, W' give the same contradiction to Lemma 4.1.

Fix $Y \in \mathcal{U}$. The adjacent pair B_1, Y has five common neighbours. Inside $\Gamma(B_1)$, the three elements of $\mathcal{U} \setminus \{Y\}$ are adjacent to Y, and B_2 is not. Thus exactly two elements of $\mathcal{M}$ are adjacent to Y. There are consequently $4 \cdot 2 = 8$ edges between $\mathcal{M}$ and $\mathcal{U}$. Since $|\mathcal{M}| = 5$, some $B_3 \in \mathcal{M}$ has at least two neighbours in $\mathcal{U}$; otherwise there would be at most five such edges.

Put $j = |\mathcal{U} \setminus \Gamma(B_3)|$, so $0 \le j \le 2$. Let h, e, ℓ count the neighbours of B_3 in $\mathcal{M} \setminus \{B_3\}$, $\mathcal{V}$ and $\mathcal{R}$, respectively. Counting common neighbours of B_1, B_3, then of B_2, B_3, and finally all neighbours of B_3, gives

$$\begin{aligned}1 + h + (4 - j) &= 5,\\ 1 + h + e &= 5,\\ 2 + h + (4 - j) + e + \ell &= 10.\end{aligned}$$

The first 1 counts B_2, the second 1 counts B_1, and the 2 counts both. Solving gives $h = j$, $e = 4 - j$ and $\ell = j$.

We will now prove $j = 0$. Define

$$\mathcal{U}_0 = \mathcal{U} \setminus \Gamma(B_3), \qquad \mathcal{V}_1 = \mathcal{V} \cap \Gamma(B_3).$$

They have sizes j and $4 - j$, and each is a clique. There are no edges between them. Indeed, for $Y \in \mathcal{U}_0$ and $W \in \mathcal{V}_1$, the five distinct blocks (B_1, B_2, B_3, Y, W) have precisely the hypotheses of Lemma 4.3: the first three form a triangle; Y is adjacent to B_1 and to neither B_2 nor B_3; and W is adjacent to B_2, B_3 and not to B_1. That lemma gives $Y \not\sim W$.

The set

$$\Omega = \{B_1, B_2, B_3\} \cup \mathcal{U}_0 \cup \mathcal{V}_1$$

is a disjoint union of size $3 + j + (4 - j) = 7$. Order its elements by listing B_1, B_2, B_3, then $\mathcal{U}_0$, then $\mathcal{V}_1$. If $j = 1$, its principal matrix C_Ω is Q_1 of Lemma 4.4; if $j = 2$, it is Q_2. To identify every entry, the first three form a triangle, every member of $\mathcal{U}_0$ is adjacent only to B_1 amongst these three, every member of $\mathcal{V}_1$ is adjacent to B_2, B_3 amongst these three, the two latter sets are separately cliques, and there are no edges between them. Equation (5) gives the displayed matrices. Both are nonsingular by Lemma 4.4, whereas Lemma 3.2 makes every seven-block principal matrix singular. Thus $j \notin \{1, 2\}$, and $j = 0$.

It follows that B_3 has no neighbours in $\mathcal{M} \setminus \{B_3\}$ and is adjacent to all four members of $\mathcal{U}$. The set $\mathcal{M} \setminus \{B_3\}$ is a clique. Otherwise two nonadjacent members, together with B_3, would be three pairwise nonadjacent neighbours of B_1. Thus the sets

$$S_1 = \{B_1, B_2\} \cup (\mathcal{M} \setminus \{B_3\}), \qquad S_2 = \{B_1, B_3\} \cup \mathcal{U}$$

are six-cliques. The disjoint partition above gives their stated intersection and union.

To prove uniqueness, let K be any six-clique containing B_1, B_2. Its four remaining members lie in $\mathcal{M}$. It cannot contain B_3, since B_3 has no neighbour in $\mathcal{M}\setminus\{B_3\}$, whereas K would have three further members there. Thus the four remaining members are exactly $\mathcal{M}\setminus\{B_3\}$, and $K = S_1$. □

Corollary 5.2. *Suppose D is a 2-$(15, 5, 2)$ design. Every adjacent pair of blocks in Z belongs to a unique six-clique. Every block belongs to exactly two six-cliques, whose intersection is that block.*

Proof. Lemma 5.1 applies to every ordered adjacent pair, so it proves the first assertion. Fix B_1. It has ten neighbours, so we may choose $B_2 \in \Gamma(B_1)$ and obtain S_1, S_2 from that lemma. Let K be a six-clique containing B_1, and choose $X \in K\setminus\{B_1\}$. The union assertion puts X in exactly one of $S_1\setminus\{B_1\}$ and $S_2\setminus\{B_1\}$. Both K and this S_i contain the adjacent pair B_1, X, so uniqueness for that pair gives $K = S_i$. Finally, S_1 and S_2 are distinct since their intersection has size one, and both have size six. □

This proves the uniqueness assertion of Hall–Connor Lemma 5.1 at $k = 5$. The explicit use of uniqueness for every adjacent pair is what lets the local constructions define a single global family.

6 The global family and the embedding: HC5

Lemma 6.1. *Suppose D is a 2-$(15, 5, 2)$ design, with block graph Z, and let $\mathcal{S}$ be the family of all six-cliques of Z. Then $|\mathcal{S}| = 7$; every block belongs to exactly two members of $\mathcal{S}$; and two distinct members of $\mathcal{S}$ have exactly one block in common. Further, for each $S_0 \in \mathcal{S}$, every point of D occurs in exactly two blocks of S_0.*

Proof. By definition every member has six blocks, and Corollary 5.2 gives exactly two members on every block. Counting incident pairs (B, S_0) gives $6|\mathcal{S}| = 2|\mathcal{B}| = 42$, whence $|\mathcal{S}| = 7$.

Two different six-cliques cannot share two different blocks: those blocks would be adjacent and would lie in two six-cliques, contrary to Corollary 5.2. To show that they always share one, fix $S_0 \in \mathcal{S}$. For each $B \in S_0$, let $f(B)$ be the other six-clique containing B. This is well-defined and lies in $\mathcal{S}\setminus\{S_0\}$. If $f(B) = f(B')$ for different $B, B' \in S_0$, then S_0 and $f(B)$ share these two blocks, which is impossible. Thus f is injective. Its domain and codomain both have size six, so it is surjective. Every other six-clique therefore shares a block with S_0, and it shares exactly one.

Fix $S_0 \in \mathcal{S}$ and put $m(p) = |\{B \in S_0 : p \in B\}|$ for $p \in P$. Counting point-block incidences in S_0, and then ordered pairs of distinct blocks of S_0 with a point in their intersection, gives

$$\sum_{p\in P} m(p) = 6 \cdot 5 = 30, \qquad \sum_{p\in P} m(p)(m(p)-1) = 6 \cdot 5 = 30.$$

The second equality uses that each such pair of blocks meets in one point. Thus, calculating in $\mathbb{Z}$,

$$\begin{aligned}\sum_{p\in P}(m(p)-2)^2 &= \sum_{p\in P} m(p)(m(p)-1) - 3\sum_{p\in P} m(p) + 4|P| \\ &= 30 - 90 + 60 = 0.\end{aligned}$$

Each summand is nonnegative, so each is zero. Thus $m(p) = 2$ for every point p. □

Theorem 6.2. [4, Theorems 2.1 and 3.2, at $k = 5$] *Suppose D is a 2-$(15, 5, 2)$ design. There is a symmetric 2-$(22, 7, 2)$ design having D as the residual design of one block. In particular, there is an integer 22×22 matrix M with $MM^T = 5I_{22} + 2J_{22}$.*

Proof. Let $\mathcal{S}$ be the seven six-cliques from Lemma 6.1. For each $S_0 \in \mathcal{S}$, adjoin a new point x_{S_0}; these seven points are distinct from one another and from every old point. Replace an old block B by

$$B^* = B \cup \{x_{S_0} : S_0 \in \mathcal{S},\ B \in S_0\},$$

and adjoin the new block $B_0 = \{x_{S_0} : S_0 \in \mathcal{S}\}$. There are $15 + 7 = 22$ points and $21 + 1 = 22$ block labels. Each old block gets exactly two new points, so every block has size seven. Each old point still occurs in seven blocks. Each new point occurs in the six old blocks of its clique and in B_0, also seven times.

Two distinct old points occur together twice, as before. An old point p and a new point x_{S_0} occur together in exactly the two blocks of S_0 containing p, by Lemma 6.1. Two distinct new points x_{S_0}, x_{S_1} occur together in the unique old block in $S_0 \cap S_1$ and in B_0, exactly twice. These are all types of pairs, so the construction is a symmetric 2-$(22, 7, 2)$ design. Deleting B_0 and its seven points recovers each original labelled block and its incidences.

We can also record the matrix identity directly. Let T be the 21×7 incidence matrix between $\mathcal{B}$ and $\mathcal{S}$, $T(B, S_0) = 1$ exactly when $B \in S_0$. The identities just proved are

$$NT = 2J_{P,\mathcal{S}}, \qquad T^T T = 5I_{\mathcal{S}} + J_{\mathcal{S}}.$$

The diagonal entries of the second product are six; its off-diagonal entries are one. With the old points first, the new points second, and the new block last, the extended incidence matrix and its product are

$$\begin{aligned} N^* &= \begin{pmatrix} N & 0 \\ T^T & \mathbf{j}_{\mathcal{S}} \end{pmatrix}, \\ N^* N^{*T} &= \begin{pmatrix} NN^T & NT \\ T^T N^T & T^T T + J_{\mathcal{S}} \end{pmatrix} \\ &= \begin{pmatrix} 5I_P + 2J_P & 2J_{P,\mathcal{S}} \\ 2J_{\mathcal{S},P} & 5I_{\mathcal{S}} + 2J_{\mathcal{S}} \end{pmatrix} = 5I + 2J. \end{aligned} \tag{8}$$

Choose arbitrary enumerations of the 22 points and 22 block labels. Reindexing N^* gives the required integer matrix M. □

This is the sufficient direction of Hall–Connor Theorem 2.1 and the existence part of their Theorem 3.2, specialized to $k = 5$. We do not need uniqueness of the embedding or the converse of Theorem 2.1.

7 The even-order obstruction: HC6

Lemma 7.1. *For $n \geq 1$ and rational numbers α, β,*

$$\det(\alpha I_n + \beta J_n) = \alpha^{n-1}(\alpha + n\beta).$$

Proof. Column i is $\alpha \mathbf{e}_i + \beta \mathbf{j}$. Expand the determinant by multilinearity in the columns. A term in which two or more columns are chosen from $\beta \mathbf{j}$ is zero, since two columns are proportional. The term using only the $\alpha \mathbf{e}_i$ columns is α^n. For each i, the term choosing $\beta \mathbf{j}$ in column i and the standard columns elsewhere is $\beta \alpha^{n-1}$. To see its sign and coefficient, expand $\mathbf{j} = \sum_{j=1}^{n} \mathbf{e}_j$ in that column. All terms except $j = i$ have two equal standard columns, and the remaining matrix is I_n. There are n choices of i, so the determinant is $\alpha^n + n\beta\alpha^{n-1} = \alpha^{n-1}(\alpha + n\beta)$. □

Lemma 7.2. *There is no integer d with $d^2 = 49 \cdot 5^{21}$.*

Proof. We will first prove that $49 \cdot 5^{2m+1}$ is not an integer square for every nonnegative integer m. Squares modulo five are $0, 1, 4$, as follows by squaring the residues $0, 1, 2, 3, 4$. In particular, if 5 divides d^2, then 5 divides d.

For $m = 0$, a proposed equality $d^2 = 49 \cdot 5$ forces $d = 5e$ for an integer e. Cancelling a factor of five gives $5e^2 = 49$, impossible modulo five. For the induction step, suppose the assertion is known for m and that $d^2 = 49 \cdot 5^{2m+3}$. Again $d = 5e$, and cancelling 25 gives $e^2 = 49 \cdot 5^{2m+1}$, contradicting the induction hypothesis. Taking $m = 10$ proves the result. □

Theorem 7.3 (The needed Bruck–Ryser–Chowla instance)**.** *There is no integer 22×22 matrix M satisfying $MM^T = 5I_{22} + 2J_{22}$.*

Proof. Take determinants, viewing the equality also over $\mathbb{Q}$. Determinant multiplicativity, invariance under transpose, and Lemma 7.1 give

$$\det(M)^2 = \det(MM^T) = \det(5I_{22} + 2J_{22}) = 5^{21}(5 + 22 \cdot 2) = 49 \cdot 5^{21}.$$

Since M has integer entries, the permutation formula for its determinant shows that $\det(M)$ is an integer. This contradicts Lemma 7.2. □

Proof of Theorem 1.1. If a 2-$(15, 5, 2)$ design existed, Theorem 6.2 would give the integer matrix excluded by Theorem 7.3. Lemma 2.1 establishes the equivalence with the five parameters in the statement. □

Theorem 7.3 is the even-order case of the obstruction usually included in the Bruck–Ryser–Chowla theorem. The proof uses neither Hasse–Minkowski nor a theorem about sums of four squares. For a symmetric 2-(v, k, λ) design the same determinant calculation gives $k^2(k - \lambda)^{v-1}$, since $k + \lambda(v - 1) = k^2$. When v is even, the odd exponent $v - 1$ forces $k - \lambda$ to be a square by prime factorization. Here we proved the necessary prime-factor obstruction by an elementary descent at five. A quadratic-form proof of the odd-order case would address a different part of the general theorem.

Corollary 7.4. *There is no design with $(v, k, b, r, \lambda) = (15, 10, 21, 14, 9)$.*

Proof. Complement each labelled block in the 15-point set. Each complement has size $15 - 10 = 5$, and each point belongs to $21 - 14 = 7$ complemented blocks. For distinct points p, q, the number of original blocks containing neither point is, by inclusion–exclusion, $21 - 14 - 14 + 9 = 2$. Thus the complement would be a 2-$(15, 5, 2)$ design, contrary to Theorem 1.1. □

8 Sources, outside facts and checks

The proof above is complete as an ordinary proof. The following table records the source of its design-theoretic steps; a source attribution does not add an unproved hypothesis.

Step here	Source	What we prove here
HC1, Lemma 2.3	Hall–Connor Lemma 4.1, quoting Connor Lemma 4.1	All intersection sizes and both multiplicities, by moments.
HC2, Lemma 2.4	Hall–Connor Lemma 4.2, quoting Connor Lemma 4.2	Both rows of the table, by $G^2 = 5G + 50J$.
HC3, Lemmas 3.1–3.2	Connor Theorem 3.1, as used by Hall–Connor	Nonnegative quadratic forms and rank at most six, including singularity above size six.
HC4, Lemmas 4.1 and 4.3	Hall–Connor Lemmas 4.3–4.5; the last two quote Connor	The claw and five-block restrictions; the first subsumes the side patterns in Remark 4.2.
HC4, Lemma 4.4	Hall–Connor (5.3), calculated there using Connor Lemma 3.1	The only two nonsingularity cases needed, by integer left inverses.
HC5, Sections 5–6	Hall–Connor Lemma 5.1, Section 6, Theorems 2.1 and 3.2	Local uniqueness, the global family, and the embedding at $k = 5$.
HC6, Section 7	Bruck–Ryser–Chowla, even-order case	The determinant formula and the obstruction at five, in full.

We did not consult Connor's paper; we use only the statements reported in Hall–Connor and prove all their required $k = 5$ instances. We do not need Connor's general determinant tool, the $k = 6$ nonexistence result, or the examples cited in the introduction of Hall–Connor. In [4], Lemmas 4.1 and 4.2 are on page 37; the local matrices (4.1), (4.4) and (4.5) are on pages 38–39. Equation (5.3) is printed on page 40 as

$$|C_{k+2}| = j(j - k + 2)(k - 6)(k + 2)^{k+1}.$$

At $k = 5$, this is $j(3 - j)7^6$, hence $2 \cdot 7^6$ for both $j = 1$ and $j = 2$. The even-order determinant argument also appears in [1, Theorem 8.3.1]. Its proof is included in Section 7 above.

There are two details worth recording for the structured proof. Hall–Connor's count of entries in their array C^* occurs before they have proved the relevant set is a clique. Our proof establishes that clique property first, then obtains exactly eight edges. Also, we explicitly define $\mathcal{S}$ to be *all* six-cliques and prove the adjacent-pair uniqueness that identifies the local choices. The two seven-block cases cannot be excluded by nonnegative principal minors alone; the rank assertion in HC3 is necessary.

The outside mathematical facts used without proof are elementary finite counting, the pigeonhole principle, matrix rank under transpose and multiplication by the transpose over $\mathbb{Q}$, the rank inequality for a zero product, rank monotonicity for submatrices, the full-rank criterion for invertibility, and basic determinant properties: multilinearity, alternation, multiplicativity and invariance under transpose. Nonnegativity of rational squares, integer division with remainder, and induction supply the arithmetic. We prove the special determinant formula and the prime obstruction in the core text. The nonnegative-minor statement is the side result in Remark 3.3; neither that remark nor Remark 4.2 is a dependency of the core proof or of its structured and Lean versions. We use no classification of strongly regular graphs, no design existence table, and no unproved theorem of Connor or Hall–Connor.

Exact numerical checks

The executable check is `checks/check_proof.py`, run from this directory by

```
python checks/check_proof.py
```

for a Python 3 with SymPy. We used Python 3.12.14 and SymPy 1.14.0. All computations use exact integers or rational numbers, and the script prints its results. The check reads the six labelled certificate matrices directly from `hallconnor.tex`, checks the two quadratic values $-2, -4$, and checks all 49 entries of each inverse identity in (7). It constructs the two seven-block patterns independently from their adjacency rules and compares every entry. It also verifies their determinants, their positive leading principal minors, and the rank-six boundary case $j = 0$.

Further checks cover the intersection counts, the algebra giving $C^2 = 35C$ and $CN^T = 0$, and the determinant

$$\det(5I_{22} + 2J_{22}) = 23365020751953125 = 49 \cdot 5^{21}.$$

An integer square-root check and division by five independently confirm the arithmetic obstruction. The script also calls `checks/check_t7.py`, which constructs the triangular graph $T(7)$ directly from the two-subsets of a seven-set. It checks both rows of the common-neighbour table, absence of claws and every five-block pattern. For all 210 ordered adjacent pairs, it checks each step of Lemma 5.1: the partition sizes $5, 4, 4, 6$, the eight edges, the choice of B_3, the counts $(h, e, \ell) = (j, 4 - j, j)$, the two six-cliques, their intersection and union, and uniqueness. It also enumerates all six-cliques and checks their incidence identities. In every local construction $j = 0$; the excluded $j = 1, 2$ matrices are checked by their separate certificates.

The biplane controls repeat the residual and reconstruction calculations for the $(11, 5, 2)$ and $(16, 6, 2)$ biplanes. Their residual block sizes are three and four, so they do not test the structural argument at block size five. No 2-$(15, 5, 2)$ design exists to test: $T(7)$ supplies a graph control but no integral incidence matrix N for the hypothetical design. The formula checks in `check_counts_and_algebra` re-evaluate the displayed algebra; they do not independently prove the counts. The controls cannot establish the quantified structural argument or the design-to-matrix bridge. The contradiction requires the integrality of N, beyond the consistent graph structure of $T(7)$.

These checks do not search over designs and do not replace any quantified argument in the proof. The largest individual certificate matrices are 7×7; the 22×22 determinant is a diagnostic check only. In Lean its value is proved symbolically, as in Lemma 7.1, not by expanding 22! terms.

What Mathlib contains

We inspected Mathlib at commit `082e2d37e8b0463410cdb532e111cd43d5a66174` (4 September 2026), the version pinned in `lake-manifest.json`, with Lean `v4.34.0-rc2`. Paths here are relative to `Mathlib/`.

- `Matrix.det_mul`, `Matrix.det_transpose` and `Matrix.det_smul`, together with `Matrix.det_reindex_self`, are in `LinearAlgebra/Matrix/Determinant/Basic.lean`. The determinant lemma `Matrix.det_one_add_replicateCol_mul_replicateRow` is in `LinearAlgebra/Matrix/SchurComplement.lean`, line 418. It states that the determinant of $I + \mathbf{u}\mathbf{v}^T$ is $1 + \mathbf{v}^T\mathbf{u}$. Scaling $5I + 2J = 5(I + (2/5)J)$ over $\mathbb{Q}$ gives the needed determinant calculation. The more general `Matrix.det_add_replicateCol_mul_replicateRow` is in the same file, line 428.

- In `LinearAlgebra/Matrix/Rank.lean`, `Matrix.rank_add_rank_le_card_of_mul_eq_zero` (line 576) gives $\operatorname{rk} C + \operatorname{rk} N^T \leq 21$ from $CN^T = 0$. `Matrix.rank_self_mul_transpose` (line 561) applies over the linearly ordered field $\mathbb{Q}$ and gives $\operatorname{rk}(NN^T) = \operatorname{rk} N$; `Matrix.rank_transpose` gives $\operatorname{rk} N^T = \operatorname{rk} N$. `Matrix.rank_of_isUnit` (line 211), with `Matrix.isUnit_iff_isUnit_det` from `LinearAlgebra/Matrix/NonsingularInverse.lean`, turns $\det(NN^T) = 5^{14} \cdot 35 \neq 0$ into $\operatorname{rk} N = 15$. Thus Lemma 7.1 at $n = 15$ and $n = 22$ serves both rank and obstruction. The file also contains `Matrix.rank_submatrix_le` and `Matrix.rank_mul_le`. No separate injectivity or `LinearMap` argument is needed. The singularity equivalence `Matrix.exists_mulVec_eq_zero_iff` is in `LinearAlgebra/Matrix/ToLinearEquiv.lean`, line 163.

- The rational quadratic-form proof can use sums of squares directly. Mathlib also has `Matrix.posSemidef_iff_dotProduct_mulVec` and `Matrix.PosSemidef.submatrix` in `LinearAlgebra/Matrix/PosDef.lean`. `Matrix.PosSemidef.det_nonneg`, in `Analysis/Matrix/PosDef.lean`, line 48, assumes an `RCLike` scalar field; it is not directly a theorem over $\mathbb{Q}$. The elementary elimination proof is retained only in Remark 3.3; it is outside the core Lean chain. The quadratic inequality and rank bound alone suffice for that chain.

- `padicValNat.mul` and `padicValNat.pow` are in `NumberTheory/Padics/PadicVal/Basic.lean`, lines 424 and 441. They give an alternative to our descent: take the natural absolute value of $\det M$, install `Fact (Nat.Prime 5)`, establish that the factors are nonzero, and deduce $2\nu_5(|\det M|) = 21$.

- In `NumberTheory/SumFourSquares.lean`, line 199, Mathlib has the precise statement

  ```
  Nat.sum_four_squares (n : Nat) :
    Exists fun a : Nat => Exists fun b : Nat =>
    Exists fun c : Nat => Exists fun d : Nat =>
      a^2 + b^2 + c^2 + d^2 = n
  ```

  Here we spell the existential quantifiers with `Exists`. The multiplicative identity `Nat.euler_four_squares` is also present. Neither declaration is a dependency of this proof. A search of the `Mathlib/` source found no declaration of Hasse–Minkowski or Bruck–Ryser–Chowla under those names. No such declaration is needed.

9 The Lean statements

We will use the structure name `Design`, as required by `cgss/notation.md`, Sections 7 and 10. This section gives the interface and the signatures that the Lean formalization, in `LineGraphs/Cited/HallConnor/`, implements, together with the plan of their proofs that we wrote before the formalization.

We will keep points and block labels as separate finite types, so that repeated blocks are allowed by the interface. One suitable shape is

```
Design (P B : Type*) [Fintype P] [Fintype B]
    [DecidableEq P] [DecidableEq B] (v k b r lam : Nat)
```

Here all five numbers `v k b r lam` are explicit parameters. The fields `card_points : Fintype.card P = v` and `card_blocks : Fintype.card B = b` link the finite types to the parameters. There is a

field `block : B -> Finset P`, a proof that every block has size `k`, a proof that every point occurs in `r` blocks, and a proof that each distinct pair occurs in `lam` blocks. It must have no injectivity field for `block`. We will use `D.incMatrix` for the integer incidence matrix, `D.compl` for the complement and `D.IsSymmetric` for the condition `v = b`, exactly as in the notation table. Lemma 2.1 converts an unbundled 2-$(15, 5, 2)$ hypothesis, without a prescribed replication number or block count, to this shape. An arbitrary 21-element label type can then be reindexed by `Fin 21`.

The following are the declaration signatures. Numeric suffixes name parameter values, not theorem numbers from the paper. We use `False` for a contradiction from a supplied design. The embedding signature returns the integer matrix and equation (8) that the obstruction consumes.

```
theorem Design.not_nonempty_15_5_2 :
  Not (Nonempty (Design (Fin 15) (Fin 21) 15 5 21 7 2))

theorem Design.false_of_parameters_15_5_2
    (D : Design P B 15 5 21 7 2) : False

theorem Design.hall_connor
    (D : Design P B 15 5 21 7 2) :
    Exists fun M : Matrix (Fin 22) (Fin 22) Int =>
      M * M.transpose =
        5 * (1 : Matrix (Fin 22) (Fin 22) Int) +
        2 * Matrix.of (fun _ _ => (1 : Int))

theorem Matrix.bruck_ryser_chowla_twenty_two
    (M : Matrix (Fin 22) (Fin 22) Int) :
    Not (M * M.transpose =
      5 * (1 : Matrix (Fin 22) (Fin 22) Int) +
      2 * Matrix.of (fun _ _ => (1 : Int)))
```

In the matrix signatures, the scalar-looking factors `5` and `2` here are scalar identity matrices under the matrix numeral convention, so their matrix products give the intended scalar multiples. Using `5` and `2` as scalar actions instead is equally suitable. For the abstract signature, the finite and decidable-equality instances for `P` and `B` are implicit section variables. The Lean formalization uses the scalar actions.

The conclusion of `Design.hall_connor` is precisely the matrix hypothesis of `Matrix.bruck_ryser_chowla_twenty_two`.

We will retain the residual assertion with `Design.hallConnorExtension`. It has point type `Sum P D.sixCliques` and block type `Option B`, where the finite set `D.sixCliques` is coerced to its subtype of members, and `none` is the new block. Its five parameters are `22 7 22 7 2`. Its incidence relation is exactly the case distinction in Theorem 6.2. Projection onto the old summands recovers `D`; this avoids making the main nonexistence theorem depend on a general residual-design API.

The internal names will describe their conclusions. Definitions use lower camel case; theorems use snake case and live in the namespace of their object. The settled incidence name `D.incMatrix` takes precedence over the paper's letter N; the other objects G, C, Z keep their letters. The parameters of a design are `v k b r lam`; `lam` avoids Lean's reserved lambda token. A concrete name map is as follows.

Object or conclusion	Declaration
Incidence and intersection matrices	`Design.incMatrix`, `Design.G`
Connor matrix and block graph	`Design.C`, `Design.Z`
Intersection restriction	`Design.inter_card_eq_one_or_two`
Degree and common neighbours	`Design.card_neighbors`, `Design.card_commonNeighbors`
Connor identities	`Design.C_mul_C`, `Design.C_mul_incMatrix_transpose`
Quadratic inequality and rank	`Design.C_quadratic_nonneg`, `Design.rank_C_le_six`
Seven-block singularity	`Design.det_C_submatrix_eq_zero`
Local restrictions	`Design.Z_clawFree`, `Design.not_adj_of_five_block_pattern`
Clique uniqueness	`Design.exists_unique_sixClique_of_adj`
Global family	`Design.sixCliques`, `Design.card_sixCliques`
Clique incidence counts	`Design.card_sixCliques_through_block`, `Design.sixClique_inter_card`, `Design.card_blocksThrough_in_sixClique`
Finite certificate values	`Computations.C19.claw_quadratic`, `Computations.C19.five_block_quadratic`, `Computations.C19.left_inverse_Q1`, `Computations.C19.left_inverse_Q2`
Complement needed by CGSS	`Design.compl`, `Design.not_nonempty_15_10_9`

Theorems in this table concerning D carry the complete hypotheses D : `Design P B 15 5 21 7 2`, as well as the blocks or cliques mentioned in their conclusions. The computation namespace is the C19 namespace already assigned in the project; the paper's lemma numbers appear in docstrings rather than in theorem names. The two named external theorems retain their names.

Formalization cost and proof boundaries

The expensive part will be the finite incidence and clique bookkeeping. In particular, Lemma 2.4 is the densest counting step: we will split the entry $(G^2)(B_1, B_2)$ into the terms $B = B_1$, $B = B_2$ and the sum over the other 19 labels, then express the three weighted sums over the four filtered finite sets counted by x_{ij}. Along with their total cardinality, these give the four displayed equations. This will require explicit exclusions, disjointness and casts from natural cardinalities to integer or rational sums. The local-to-global clique argument will also need disjoint partitions, double counts with block labels and exact filtered-set cardinalities.

The matrix rank lemmas listed above make the rank deduction short. The remaining transport cost is the principal-submatrix bound and an explicit equivalence from a seven-block subset to `Fin 7`, so that the entry pattern and singularity agree with the certificates.

The passage from Theorem 6.2 to Theorem 7.3 has a separate design-to-matrix cost. If the embedding returns a bundled design, we must apply DS1, $NN^T = (r - \lambda)I + \lambda J$, to its incidence matrix, and reindex the point and block types to `Fin 22`. A determinant taken on the original point type must be transported by `Matrix.det_reindex_self`. Our chosen `Design.hall_connor` signature instead returns the matrix identity directly: we will prove the block product (8) and reindex its rows and columns to `Fin 22`. This still requires compatibility of reindexing with multiplication,

transpose, identity and all-ones matrices; it avoids making the contradiction depend on a second application of DS1 to a bundled extension.

The small certificate identities should be inexpensive. We will use kernel-checked finite matrix multiplication, rational arithmetic, and linear combinations of rows. The moment polynomials must be evaluated in $\mathbb{Z}$ or $\mathbb{Q}$, not with truncated subtraction in $\mathbb{N}$. The determinant obstruction should also be inexpensive once the general rank-one determinant identity is instantiated; either the descent of Lemma 7.2 or the existing valuation lemmas will finish it. We will keep N^* integral and cast it to $\mathbb{Q}$ only for the determinant identity.

There is no unresolved mathematical step in this proof. The structured proof is `cgss/proofs/hallconnor/hallconnor-lamport.tex`. The Lean formalization, in the folder `LineGraphs/Cited/HallConnor/`, implements the `Design` interface and these declarations. Neither needs a quadratic-form local-global principle or a development of sums of four squares.

References

[1] A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Cambridge University Press, 2022.

[2] P. J. Cameron, J. M. Goethals, J. J. Seidel and E. E. Shult, *Line graphs, root systems, and elliptic geometry*, J. Algebra **43** (1976), 305–327.

[3] W. S. Connor, Jr., *On the structure of balanced incomplete block designs*, Ann. Math. Statist. **23** (1952), 57–71. We use the statements reproduced in [4], with proofs given above.

[4] M. Hall, Jr. and W. S. Connor, *An embedding theorem for balanced incomplete block designs*, Canad. J. Math. **6** (1954), 35–41. doi:10.4153/CJM-1954-005-4.